Closed-Loop Rolling Calibration for Forecast-Informed Reservoir Dispatch

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***Index Terms*—reservoir inflow forecasting, rolling calibration, data assimilation, forecast-informed operation, model predictive control**

***Abstract*—Real-time reservoir dispatch depends critically on short-horizon inflow forecasts, yet nonstationarity and shifting operational constraints routinely erode calibration quality and degrade forecast-informed decision-making. We present a closed-loop framework that tightly couples rolling model calibration, uncertainty-aware probabilistic forecasting, and implementation feedback within a receding-horizon dispatch loop. At each decision cycle, model parameters are updated from the most recent observations; a probabilistic multi-step inflow forecast is generated; a scenario-based receding-horizon dispatch problem is solved; and realized outcomes are used to adaptively inflate forecast uncertainty when recent errors grow. We evaluate the framework on three reservoirs from the ResOpsUS historical operations dataset (daily storage, inflow, and outflow records spanning 1980–2020), using a strictly causal train/tune/test partition (1980–2010 / 2011–2014 / 2015–2020) and a 7-day forecast and dispatch horizon. Against a statically calibrated AR(p) baseline, rolling calibration achieves comparable point-forecast skill while reducing mean CRPS by 16.3% across reservoirs. In end-to-end dispatch simulation, closed-loop scenario MPC with feedback attains the lowest mean objective among all tested pipelines, reducing the mean operational objective by 0.33% relative to a static open-loop deterministic MPC baseline; performance gains are reservoir-dependent, storage-bound violation rates remain comparable across methods, and the full decision cycle executes in approximately 20 ms per day on CPU.**

I. Introduction

Real-time reservoir operation relies on short-term inflow predictions to schedule releases while satisfying safety constraints (e.g., flood control limits) and operational objectives (e.g., hydropower generation, water supply). Classical reservoir operation studies emphasize that optimal policies depend on both hydrologic dynamics and decision constraints [1], [2].

Forecast errors and nonstationary conditions can induce systematic bias and drift in inflow predictions. This drift raises operational risk. Yet the mapping from forecast skill to operational value is complex and regime-dependent [3], [4].

Rolling calibration and data assimilation provide a principled mechanism to adapt model states and parameters online using incoming observations. Ensemble Kalman filter (EnKF) variants have been widely used for dual state–parameter estimation and for improving streamflow forecasting [5]–[8]. The inflow forecasting literature also spans conceptual models, machine learning baselines, and uncertainty-aware multi-step prediction [9]–[12].

Forecast-informed reservoir operation (FIRO-style) and model predictive control (MPC) approaches integrate forecasts into dispatch decisions, including uncertainty-aware optimization using scenarios or ensembles [13]–[16]. However, many existing workflows treat forecasting and dispatch as a sequential, largely open-loop pipeline: forecasts are produced, an operating policy is computed, and the system is executed with limited explicit coupling of implementation outcomes back into the next forecast-and-dispatch cycle.

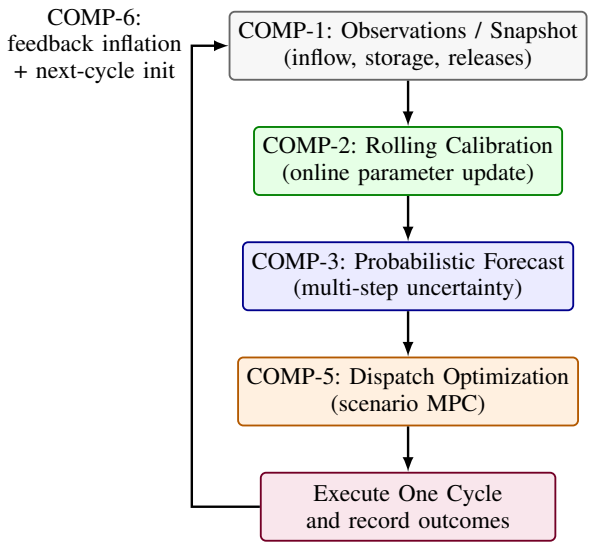

Fig. 1. Closed-loop coupling of rolling calibration, forecasting, dispatch, and implementation feedback.

This paper proposes a closed-loop method that couples (i) rolling parameter calibration for inflow forecasting, (ii) uncertainty-aware dispatch optimization, and (iii) implementation feedback that updates the next decision cycle. We test whether making this coupling explicit improves operational reliability under nonstationarity relative to open-loop baselines with static calibration or without uncertainty propagation.

A. Contributions

C1 Closed-loop forecast–dispatch–feedback framework. We propose a unified closed-loop architecture that couples inflow forecasting, scenario-based dispatch optimization, and implementation feedback. Evaluated on ResOpsUS (3 reservoirs; test period 2015–2020), closed-loop scenario MPC with feedback reduces the mean operational objective by 0.33% relative to a static open-loop deterministic MPC baseline, with reservoir-dependent gains and comparable storage-bound violation rates.

TABLE I
TAXONOMY OF BUILDING BLOCKS FOR FORECAST-INFORMED RESERVOIR OPERATION.

Building block	Representative references
Rolling calibration / data assimilation	[5]–[8]
Multi-step inflow forecasting	[9]–[12]
Forecast-informed operation and FIRO-style methods	[16], [17]
Uncertainty-aware dispatch (scenario MPC)	[13]–[15]
Forecast skill vs operational value	[3]

C2 Rolling calibration with quantified probabilistic skill. We instantiate online model updating via a lightweight AR(p) inflow forecaster and assess both point and probabilistic forecast accuracy. Rolling calibration reduces mean CRPS by 16.3% relative to a static AR(p) baseline while leaving RMSE essentially unchanged, demonstrating that distributional accuracy improves without sacrificing point-forecast fidelity.

C3 Ablation of uncertainty propagation and feedback coupling. We systematically isolate the contribution of each pipeline component. Removing implementation feedback increases the operational objective by 0.16%; replacing scenario MPC with deterministic dispatch increases it by 0.37%.

C4 Stress tests and computational characterization. Scenario-based closed-loop optimization executes in approximately 20 ms per daily cycle on CPU. The framework remains stable under synthetic regime shifts and sparse or delayed observations, with objective variation below 0.3%, confirming practical robustness without incurring prohibitive computational overhead.

II. RELATED WORK

We organize related work around the building blocks of a forecast–dispatch closed loop: (i) rolling calibration, (ii) inflow forecasting with uncertainty, and (iii) forecast-informed dispatch under constraints.

A. Rolling Calibration and Data Assimilation

EnKF-based dual estimation and related data assimilation methods update hidden states and time-varying parameters online to reduce forecast errors [5], [6]. Recent work also emphasizes the role of time-varying parameters under changing environments and the need to constrain filtering behavior for stable performance [7], [8].

B. Inflow Forecasting and Uncertainty

Reservoir inflow forecasting spans complementary modeling frameworks and machine learning baselines, increasingly with uncertainty quantification for multi-step prediction [9]–[12]. For operational use, uncertainty calibration (e.g., CRPS and coverage) is often as important as point-skill metrics [18].

C. Forecast-Informed Reservoir Operation

Forecast-informed operation integrates inflow predictions into dispatch policies, including data-driven and recurrent-network approaches [16], [17], [19]. Yet improved forecast skill does not guarantee proportional operational gains, which motivates evaluating both forecast metrics and operational outcomes [3]. Recent studies also combine multi-step streamflow forecasts with multi-objective reservoir operation and performance assessment [20].

D. Uncertainty-Aware Dispatch and MPC

Scenario-based MPC provides a natural interface between probabilistic forecasts and constraint-aware reservoir dispatch [14], [15]. Domain-specific studies also show that forecast uncertainty varies across hydrologic periods, suggesting that operating rules should account for these shifts [13]. MPC-style operation has also been explored for flood mitigation in multi-reservoir settings and for ensemble-forecast-driven multi-objective dam operation [21], [22].

III. METHOD

We describe a modular closed-loop framework that couples rolling forecast calibration with forecast-informed dispatch. The design follows a receding-horizon decision cycle and can wrap either conceptual or data-driven inflow models. We emphasize the coupling logic rather than introducing a new forecasting architecture.

A. Problem Setup

Let k index decision cycles with control interval Δt (e.g., hourly or daily). At the beginning of cycle k, the operator observes a state snapshot x_k (e.g., storage, recent inflows, releases, and constraint indicators) and optionally exogenous covariates (e.g., meteorological forcings). The task is to compute a release plan over a finite horizon H while satisfying operational constraints (e.g., storage bounds, release limits, and safety constraints) [1], [2].

We distinguish between:

- **Forecasting**: generate a probabilistic inflow forecast $\hat{q}_{k+1:k+H}$.
- **Dispatch**: compute releases $r_{k:k+H}$ (or a subset such as r_k) using the forecast.
- **Implementation**: apply the release for one cycle, observe outcomes, and feed them back.

a) Notation: We use q_t for realized inflow at day t, $\hat{q}_{t+1:t+H}$ for a horizon-H inflow forecast, r_t for the implemented release, and s_t for storage. Subscripts k and t both index decision cycles (we use t for day-indexed equations consistent with the implementation).

B. Approach Overview

The framework consists of six components: (i) observation/state snapshot, (ii) rolling parameter calibration, (iii) probabilistic inflow forecasting, (iv) scenario generation, (v) scenario-based MPC dispatch, and (vi) an implementation-feedback loop orchestrator.

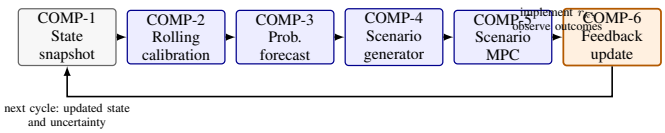

Fig. 2. Method pipeline components. After scenario MPC selects the first control action, realized outcomes enter COMP-6, which updates next-cycle state and uncertainty information for the following decision cycle.

Algorithm 1 Closed-loop forecast–dispatch–implementation coupling (per cycle k).

1: Observe state snapshot x_k and recent operations/measurements.
2: Update forecast-model state/parameters via rolling calibration: $\theta_k \leftarrow \mathrm{Calibrate}(x_k)$.
3: Generate probabilistic inflow forecast over horizon H: $\hat{q}_{k+1:k+H} \leftarrow \mathrm{Forecast}(x_k, \theta_k)$.
4: Convert forecasts to scenarios (optional): $\mathcal{S}_k \leftarrow \mathrm{Scenarios}(\hat{q}_{k+1:k+H})$.
5: Solve uncertainty-aware dispatch (scenario MPC): $r_{k:k+H} \leftarrow \mathrm{Dispatch}(x_k, \mathcal{S}_k)$.
6: Implement r_k for one cycle; observe realized outcomes and constraint status.
7: Advance to cycle $k{+}1$ using implementation feedback.

a) Rolling parameter calibration: We update model state and parameters online using incoming observations, motivated by EnKF dual-estimation and constrained filtering for stable forecasting [5]–[8]. The calibration step outputs an updated parameter set θ_k and (optionally) an updated state estimate used by the forecaster.

b) Probabilistic inflow forecasting and scenario generation: Given θ_k and the current state, the forecaster produces an ensemble (or quantiles) of inflow trajectories. Scenarios are passed to the dispatcher either directly (ensemble-as-scenarios) or via a lightweight scenario generator. We evaluate uncertainty calibration using proper scoring rules such as CRPS [18].

c) Uncertainty-aware dispatch (scenario MPC): The dispatcher solves a receding-horizon optimization that accounts for constraints and forecast uncertainty. Scenario-based MPC provides a natural interface for this step [14], [15], [23].

d) Implementation feedback and closed-loop coupling: After executing the first control action (or first segment) of the plan, we observe realized inflow, state transitions, and constraint outcomes. These signals are injected into the next cycle. This closes the loop between forecast, dispatch, and implementation, enabling ablations that isolate the value of rolling calibration, uncertainty propagation, and feedback coupling.

C. Closed-Loop Cycle Algorithm

D. Rolling Calibration via Recursive Least Squares

For clarity and reproducibility, we instantiate the rolling-calibration module (COMP-2) using a lightweight autoregressive forecaster with online parameter updates. Let q_t denote daily inflow at day t. We use an AR(p) model with intercept:

$$q_t = \beta_0 + \sum_{i=1}^{p} \beta_i q_{t-i} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_t^2). \tag{1}$$

Define the feature vector $\phi_t = [1, q_{t-1}, \ldots, q_{t-p}]^\top$ and parameter vector $\theta_t = [\beta_0, \beta_1, \ldots, \beta_p]^\top$. Rolling calibration updates θ_t using recursive least squares (RLS) with exponential forgetting factor λ:

$$K_t = \frac{P_{t-1}\phi_t}{\lambda + \phi_t^\top P_{t-1}\phi_t}, \tag{2}$$

$$\theta_t = \theta_{t-1} + K_t \left(q_t - \phi_t^\top \theta_{t-1}\right), \tag{3}$$

$$P_t = \frac{1}{\lambda}\left(P_{t-1} - K_t \phi_t^\top P_{t-1}\right). \tag{4}$$

We initialize θ using a static ridge fit on the training period and warm-start RLS through the tuning period so the test period begins from a stable state. We track predictive uncertainty with an EWMA variance estimator:

$$\sigma_t^2 = \alpha \sigma_{t-1}^2 + (1-\alpha)\left(q_t - \phi_t^\top \theta_{t-1}\right)^2, \tag{5}$$

where $\alpha \in [0, 1)$ controls adaptation speed.

E. Probabilistic Forecast and Scenario Generation

Given θ_t and history up to day $t{-}1$, we produce a multi-step mean forecast $\mu_{t+1:t+H}$ by iterating the AR recursion forward. We approximate the h-step predictive standard deviation by $\sigma_t\sqrt{h}$, reflecting accumulation of one-step residual uncertainty over the horizon. For scenario MPC, we generate $|\mathcal{S}|$ inflow scenarios by sampling i.i.d. standard Gaussians $z_h^{(s)}$ and forming:

$$q_{t+h}^{(s)} = \max\left(0,\, \mu_{t+h} + \gamma_t\, \sigma_t \sqrt{h}\, z_h^{(s)}\right), \tag{6}$$

where γ_t is an optional feedback inflation factor (COMP-6). Deterministic dispatch is recovered by setting $|\mathcal{S}|{=}1$ and $z_h^{(1)}{=}0$.

F. Scenario MPC Dispatch Objective

For each scenario, we simulate the storage dynamics over the horizon and solve a receding-horizon optimization for the release sequence. The stage cost is a quadratic proxy that trades off target tracking, actuation effort, smoothness, and soft penalties for storage violations:

$$\begin{aligned}\ell(s_{t+1}, r_t, r_{t-1}) = {} & w_s(s_{t+1} - s^\star)^2 + w_r r_t^2 \\ & + w_{\Delta r}(r_t - r_{t-1})^2 + w_v\, \mathrm{Viol}(s_{t+1}).\end{aligned} \tag{7}$$

We define $\mathrm{Viol}(s) = \max(0, s - \bar{s})^2 + \max(0, \underline{s} - s)^2$ and constrain releases to $r_t \in [\underline{r}, \bar{r}]$. The MPC implements only the first release each cycle, then replans at the next cycle using updated observations and calibrated parameters.

G. Implementation Feedback via Adaptive Inflation

Closed-loop coupling (COMP-6) modifies the next-cycle scenario spread based on recent realized errors. We maintain a fast EWMA of squared one-step prediction errors e_t^2 and compute:

$$\gamma_t = \mathrm{clip}\left(\sqrt{\frac{\mathrm{EWMA}(e_t^2)}{\sigma_t^2}},\, 1,\, \gamma_{\max}\right), \tag{8}$$

so scenario dispersion increases when short-term errors spike. This is a simple, implementable feedback mechanism that can be ablated independently from rolling calibration and uncertainty propagation.

IV. Experimental Setup

A. Datasets and Preprocessing

We evaluate on ResOpsUS, a dataset of historical reservoir operations in the contiguous United States [24]. Reservoir metadata are linked to global dam inventories such as GRanD [25]. We select three reservoirs with complete daily records from 1980-01-01 to 2020-12-31: Glen Canyon (DAM_ID 597, AZ), Flaming Gorge (DAM_ID 451, UT), and Navajo (DAM_ID 601, NM). Each day includes storage (MCM), inflow (m^3/s), and outflow (m^3/s).

We use a strictly chronological split: training (1980–2010), tuning (2011–2014), and test (2015–2020), reflecting nonstationarity and regime shifts [4]. All rolling calibration and dispatch decisions in the test period are strictly causal: the decision for day t uses information available up to day $t-1$. Figure 3 visualizes the daily inflow series and split boundaries for the three case studies.

B. Closed-Loop Simulator and Constraints

Closed-loop operational performance is evaluated by simulating the reservoir mass balance and operational constraints. Following standard reservoir optimization formulations [1], [2], we use a discrete-time storage update:

$$s_{k+1} = s_k + (q_k - r_k - \mathrm{spill}_k)\Delta t, \tag{9}$$

where s_k is storage, q_k is (realized) inflow, and r_k is the implemented release. Constraints include storage bounds, release bounds, and optional ramping/operational constraints. For the ResOpsUS case studies, storage and release bounds are derived from training-period quantiles. Over multi-year rollouts, unmodeled losses (e.g., evaporation or spill not available in the time series) can cause drift. We estimate an aggregate daily loss term from training mass-balance residuals and treat it as constant during simulation. The dispatch module is solved in a receding-horizon model predictive control (MPC) manner [15]. Uncertainty-aware variants use scenarios generated from forecast distributions or ensembles [13], [14].

C. Rolling-Calibration Evaluation Protocol

We report both forecast skill and closed-loop operational value, since improvements in forecast accuracy do not necessarily translate into proportional operational gains [3]. For each decision cycle in the test period, we:

- update forecast-model parameters using a rolling window of the most recent W observations, using either EnKF-style dual state–parameter estimation when applicable [5], [6] or a constrained sliding-window optimizer;
- generate an inflow forecast over horizon H (optionally probabilistic), and map uncertainty to a scenario set for the dispatch optimizer;
- solve the dispatch problem over horizon H and implement only the first control action;
- advance the simulator using realized inflows, and record objective and constraint outcomes.

To reduce identifiability artifacts in online estimation, time-varying parameters are regularized and bounded, and we monitor stability diagnostics during rollouts [7], [8].

D. Baselines

Baselines are grouped by forecasting and operation components. The main comparisons follow a controlled-ablation philosophy:

- Forecast baselines: persistence, seasonal climatology (day-of-year mean), a statically calibrated AR(p) time-series model, and an ML-LSTM sequence model [10], [11]. Our forecaster uses the same AR(p) structure with online rolling calibration and probabilistic error modeling.
- Operation baselines: static-calibration open-loop deterministic MPC, rolling-calibration open-loop deterministic MPC, rolling-calibration open-loop scenario MPC, and the proposed rolling-calibration closed-loop scenario MPC with feedback.

E. Metrics

We report both forecast skill and operational performance metrics:

- Forecast metrics: RMSE/MAE, Nash–Sutcliffe efficiency (NSE) [26], and uncertainty metrics such as CRPS and coverage [18].
- Operational metrics: constraint-violation rate and objective value, plus reliability/resiliency/vulnerability-style criteria when appropriate [27].

F. Implementation Details and Reproducibility

Unless otherwise stated, all methods are evaluated at the same decision cycle Δt and horizon H under identical data splits and pre-processing. For stochastic baselines (e.g., neural forecasting models), we report results across multiple random seeds and document training and early-stopping criteria on the tuning period. For each run, we report the rolling calibration configuration (window length W, update frequency, and any regularization), the scenario generation settings (scenario count and sampling scheme), and the dispatch solver

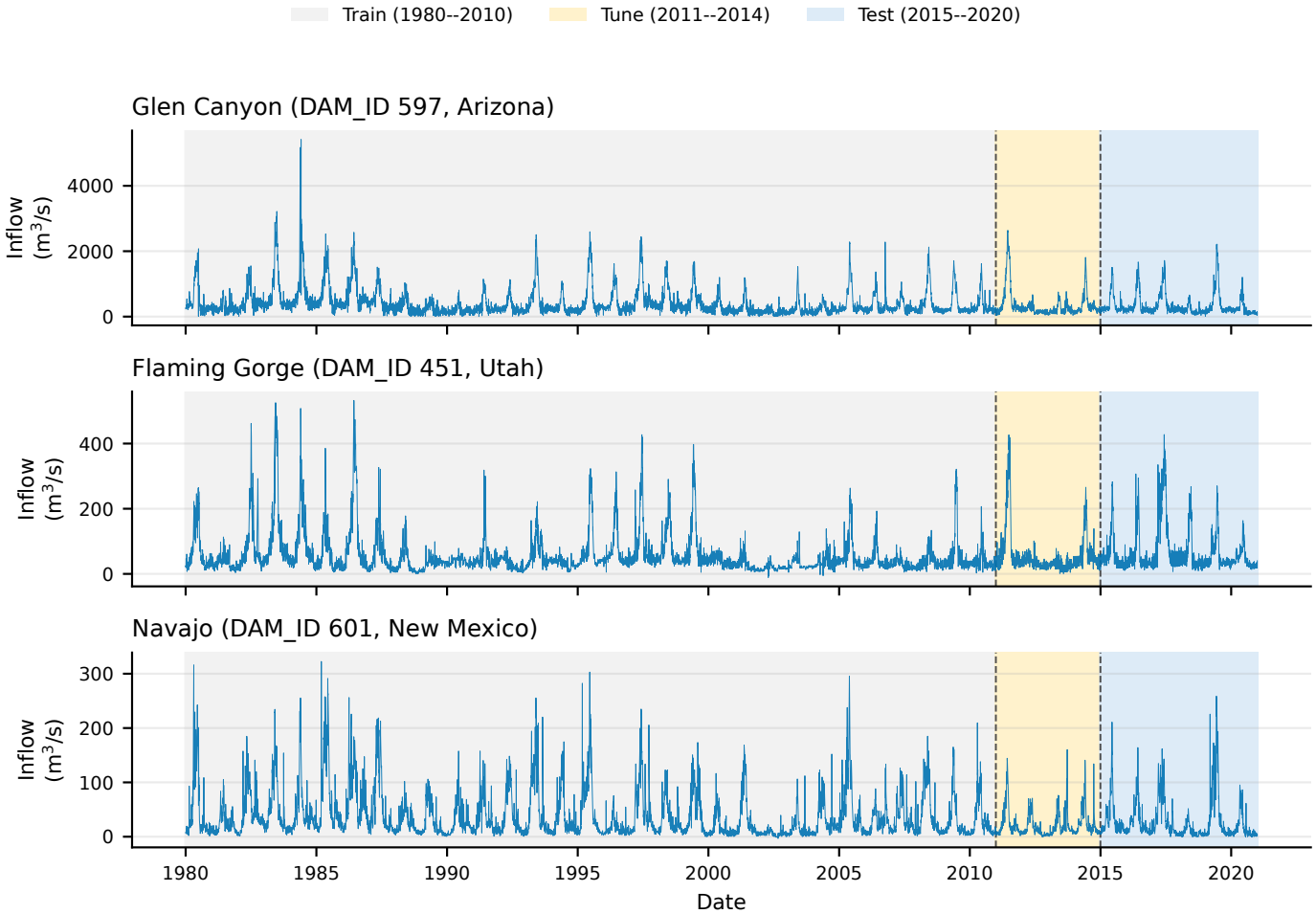

Fig. 3. Daily inflow time series for the three ResOpsUS reservoirs (1980–2020) with chronological splits (train/tune/test).

TABLE II
EXPERIMENTAL SETUP SUMMARY (RESOPSUS CASE STUDIES).

Item	Specification
Case study	ResOpsUS (3 reservoirs; daily 1980–2020; test 2015–2020)
Decision cycle / horizon	Δt = 1 day; H = 7 days
Rolling calibration	AR(3) recursive least squares; forgetting factor λ=0.999; update freq.=1/day
Uncertainty propagation	Gaussian residual model; EWMA variance (α=0.98); $\lvert\mathcal{S}\rvert$=20 scenarios; feedback inflation max=3.0
Bounds	storage/release bounds from training 5%–95% quantiles
Forecast metrics	RMSE, MAE, NSE; CRPS; 80% coverage
Operational metrics	storage-bound violation rate; objective; reliability/vulnerability

settings (tolerances and maximum iterations). We also report wall-clock runtime per decision cycle to assess operational feasibility.

TABLE III
FORECAST MAIN RESULTS ON RESOPSUS (TEST 2015–2020; H=7 DAYS).

Method	RMSE	NSE	CRPS	Cov.
Persistence	54.44±55.82	0.836±0.044	34.87±40.41	0.940±0.036
Climatology	102.81±110.70	0.473±0.088	92.66±109.10	0.979±0.013
AR(p) static	53.33±55.90	0.848±0.039	31.11±35.21	0.937±0.026
ML-LSTM	57.63±62.10	0.834±0.031	28.89±31.93	0.901±0.019
Rolling AR(p) (ours)	53.40±55.45	0.845±0.044	26.03±27.79	0.915±0.020

V. RESULTS

We report forecast skill and operational value on the ResOpsUS case studies. Unless stated otherwise, table entries are mean±std across the three reservoirs in the test period (2015–2020).

A. *Forecast Skill*

We report multi-step inflow forecast skill over the operational horizon using point metrics (RMSE/MAE and NSE [26]). When probabilistic forecasts are available, we also report proper scoring rules such as CRPS and empirical coverage [18]. To reflect nonstationarity, we stratify test performance by hydrologic regime when possible [4].

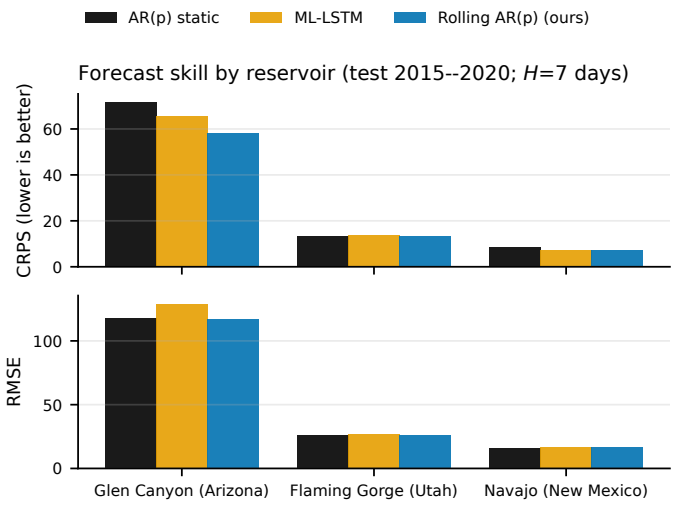

Fig. 4. Forecast skill broken down by reservoir for key methods (test 2015–2020; H=7 days).

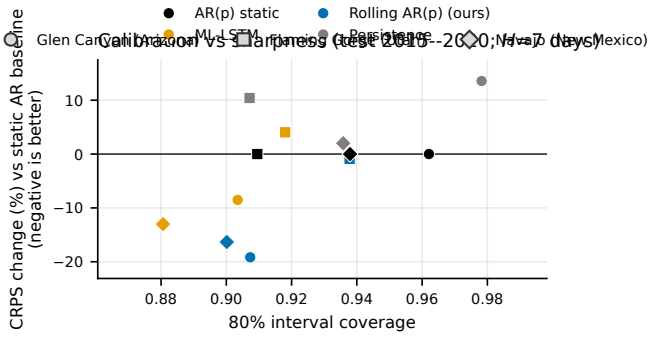

Fig. 5. Calibration vs sharpness tradeoff for key forecasters (test 2015–2020; H=7 days). X-axis: nominal 80% interval coverage. Y-axis: CRPS change (%) relative to the static AR baseline for the same reservoir (negative is better). Marker shape indicates reservoir; dashed vertical line marks the nominal 0.8 coverage.

Rolling calibration primarily improves probabilistic calibration. Relative to the static AR(p) baseline, mean CRPS decreases from 31.11 to 26.03 (16.3%). RMSE remains essentially unchanged (53.33 to 53.40; $< 0.2\%$ increase), and NSE shifts from 0.848 to 0.845. Coverage for the nominal 80% interval decreases from 0.937 to 0.915, indicating less conservative uncertainty while improving CRPS. Figure 5 shows that all evaluated forecasters remain conservative (coverage > 0.8), but rolling calibration reduces over-coverage while lowering CRPS. Figure 4 shows that this improvement is heterogeneous across reservoirs: relative to the static AR(p) baseline, CRPS decreases by 0.9% (Flaming Gorge), 19.2% (Glen Canyon), and 16.3% (Navajo).

B. Operational Main Results

We report end-to-end closed-loop performance under receding-horizon dispatch [15]. We include both deterministic and scenario-based variants [13], [14]. We focus on constraint violations, objective value, and reliability-style criteria [27]. We analyze operational value separately from forecast skill [3].

TABLE IV
OPERATIONAL MAIN RESULTS ON RESOPSUS (TEST 2015–2020). OBJECTIVE IS REPORTED IN $\times 10^6$ COST UNITS (LOWER IS BETTER).

Method	Viol.	Obj. ($\times 10^6$)	Notes
Static det. MPC (baseline)	0.032±0.056	9.20±8.28	open-loop + static params
Rolling det. MPC	0.032±0.056	9.21±8.28	isolates rolling calibration
Rolling scenario MPC	0.032±0.056	9.19±8.26	isolates uncertainty-aware dispatch
Closed-loop scenario MPC (ours)	0.032±0.056	9.17±8.24	closed-loop coupling

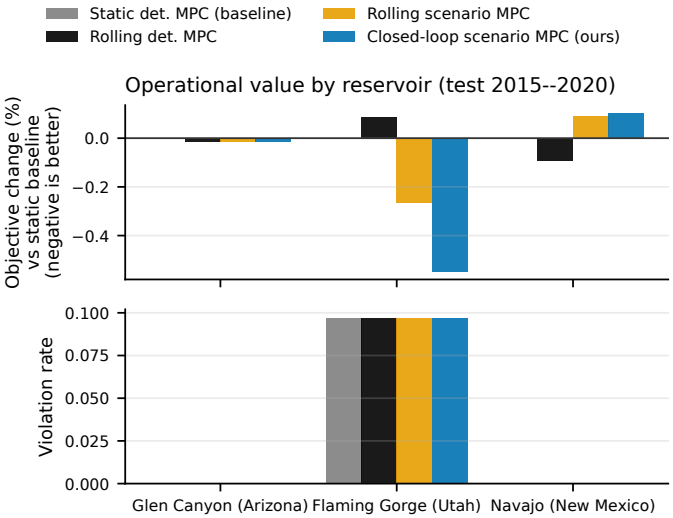

Fig. 6. Operational objective change (relative to static open-loop deterministic MPC) and storage-bound violation rate, broken down by reservoir (test 2015–2020). Objective change is reported as a percent difference; negative values indicate lower cost (better).

Closed-loop scenario MPC with feedback achieves the lowest average objective among the tested pipelines, with similar storage-bound violation rates.

Compared with the static open-loop deterministic baseline, closed-loop scenario MPC with feedback reduces the mean operational objective from 9.20 to 9.17×10^6 (0.33%). The mean violation rate is similar (0.0324 vs. 0.0322). Violations are concentrated in Flaming Gorge, while the other two reservoirs have zero violations across all tested pipelines (Figure 6). The objective change is reservoir-dependent: relative to the static baseline, our method changes objective by -0.55% (Flaming Gorge), -0.02% (Glen Canyon), and $+0.10\%$ (Navajo).

C. Ablation Studies

We ablate the core coupling components to isolate their marginal value (EXP-ABL-*): rolling calibration, uncertainty propagation (scenario-based dispatch), and implementation feedback.

The ablation results attribute most objective gains to uncertainty propagation and feedback (Figure 7). Relative to the full method (9.17×10^6), removing feedback increases the objective to 9.19×10^6 (+0.16%). Replacing scenario MPC

TABLE V
ABLATION RESULTS ON RESOPSUS (TEST 2015–2020). OBJECTIVE IS REPORTED IN $\times 10^6$ COST UNITS (LOWER IS BETTER).

Method	Viol.	Obj. ($\times 10^6$)	Notes
Full closed-loop method	0.032±0.056	9.17±8.24	reference
w/o rolling calibration (static params)	0.032±0.056	9.17±8.23	ablate COMP-2
w/o implementation feedback (open loop)	0.032±0.056	9.19±8.26	ablate COMP-6
Deterministic dispatch (no scenarios)	0.032±0.056	9.21±8.28	ablate uncertainty propagation
Point forecast only (no UQ)	0.032±0.056	9.21±8.28	remove probabilistic forecasts

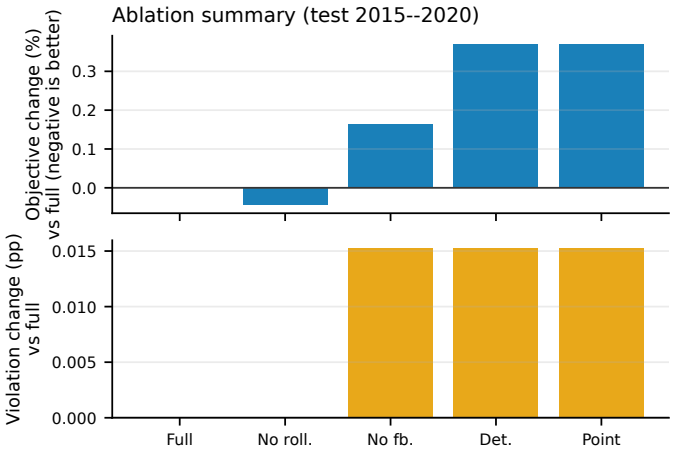

Fig. 7. Ablation summary relative to the full closed-loop method (test 2015–2020). Top: objective change (%) vs full; negative indicates lower cost (better). Bottom: storage-bound violation-rate change in percentage points (pp) vs full.

with deterministic dispatch (or using point forecasts only) increases it to 9.21×10^6 (+0.37%). In this proxy objective, removing rolling calibration changes the objective by less than 0.05%.

D. Error Analysis / Robustness

We evaluate robustness to synthetic regime shifts (multiplicative inflow scaling) and to sparse/delayed observations used for rolling calibration (EXP-ROB-*). Stress tests are motivated by known failure modes in online data assimilation and calibration [5]–[7].

a) Inflow regime shift.: Under severe inflow scaling (α=1.2), closed-loop feedback reduces the mean objective from 25.73 to 25.64×10^6 (0.36%) while maintaining essentially identical violation rates (0.0677). This result suggests that the feedback mechanism can provide a modest advantage over the open-loop baseline under a substantial shift in inflow magnitude.

b) Sparse and delayed observations.: When rolling calibration operates under degraded observational conditions—up to 50% missing measurements and a 3-day reporting delay—the mean objective varies by less than 0.3% relative to the nominal setting, and violation rates remain tightly bounded within [0.03224, 0.03239]. This insensitivity indicates that the calibration loop can tolerate realistic monitoring imperfections without large operational degradation.

TABLE VI
ROBUSTNESS STRESS TESTS (EXCERPT). OBJECTIVE IS REPORTED IN $\times 10^6$ COST UNITS (LOWER IS BETTER). FULL RESULTS ARE PROVIDED IN THE ACCOMPANYING ARTIFACTS.

Stress	Severity	Method	Viol.	Obj. ($\times 10^6$)
inflow_scale	0.8	Static det. MPC (baseline)	0.014	7.74
inflow_scale	0.8	Rolling det. MPC	0.014	7.74
inflow_scale	0.8	Closed-loop scenario MPC (ours)	0.014	7.74
inflow_scale	0.9	Static det. MPC (baseline)	0.023	6.87
inflow_scale	0.9	Rolling det. MPC	0.023	6.88
inflow_scale	0.9	Closed-loop scenario MPC (ours)	0.023	6.87
inflow_scale	1.0	Static det. MPC (baseline)	0.032	9.20
inflow_scale	1.0	Rolling det. MPC	0.032	9.21
inflow_scale	1.0	Closed-loop scenario MPC (ours)	0.032	9.17
inflow_scale	1.1	Static det. MPC (baseline)	0.042	15.15
inflow_scale	1.1	Rolling det. MPC	0.042	15.16
inflow_scale	1.1	Closed-loop scenario MPC (ours)	0.042	15.13
inflow_scale	1.2	Static det. MPC (baseline)	0.068	25.73
inflow_scale	1.2	Rolling det. MPC	0.068	25.74
inflow_scale	1.2	Closed-loop scenario MPC (ours)	0.068	25.64
obs_sparse_delay	miss=0.0,delay=0	Closed-loop (ours)	0.032	9.18
obs_sparse_delay	miss=0.0,delay=1	Closed-loop (ours)	0.032	9.19
obs_sparse_delay	miss=0.0,delay=3	Closed-loop (ours)	0.032	9.19

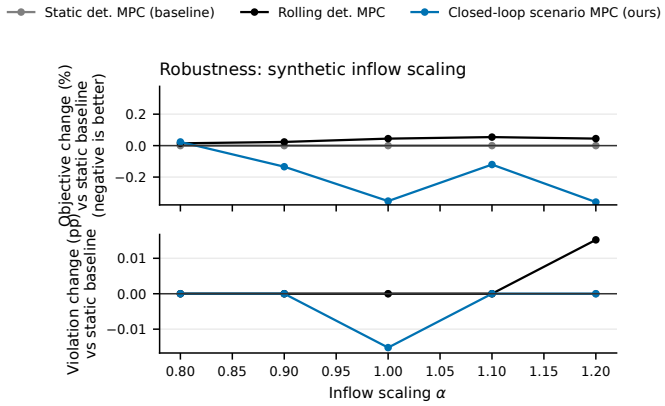

Fig. 8. Robustness to a synthetic regime shift induced by multiplicative inflow scaling (α). Top: objective change (%) vs static open-loop deterministic MPC (baseline) at each α; negative indicates lower cost (better). Bottom: storage-bound violation-rate change in percentage points (pp) vs baseline.

E. Efficiency

We measure per-cycle runtime for calibration and dispatch to assess feasibility at a daily decision cadence (EXP-EFF-RUNTIME).

Scenario-based MPC requires about 20 ms per daily cycle on CPU, compared with about 3.3 ms for the deterministic baseline. Rolling-calibration updates contribute roughly 0.04 ms per cycle, so runtime is dominated by the dispatch optimization.

VI. DISCUSSION

A. Forecast Skill vs. Operational Value

Rolling calibration leaves point-skill nearly unchanged but improves probabilistic calibration (CRPS). This pattern is

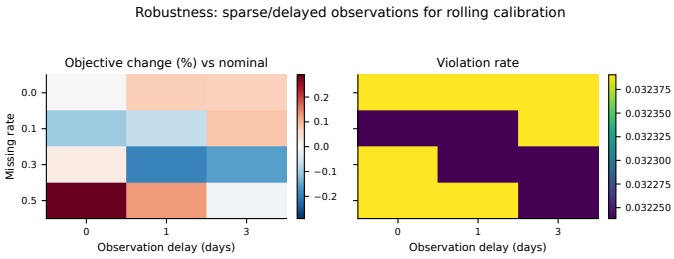

Fig. 9. Robustness of closed-loop rolling calibration to sparse and delayed observations. Left: objective change relative to nominal (miss $= 0$, delay=0); negative indicates lower cost (better). Right: violation rate.

TABLE VII
EFFICIENCY COMPARISON (MEAN PER-CYCLE RUNTIME ON CPU).

Method	Calib (s)	Dispatch (s)	Total (s)
Static det. MPC (baseline)	0.000	0.003	0.003
Rolling scenario MPC	0.000	0.020	0.020
Closed-loop scenario MPC (ours)	0.000	0.020	0.020

consistent with the idea that operational value depends on the decision context, not only on point accuracy [3]. In our proxy objective, closed-loop scenario MPC yields a modest average improvement (0.33%), while constraint violation rates remain comparable across pipelines. The per-reservoir breakdown (Figures 4 and 6) suggests that the operational gain is driven primarily by a subset of reservoirs and periods where uncertainty interacts with storage bounds. Figure 5 further indicates that all evaluated forecasters remain conservative (coverage > 0.8), and that rolling calibration reduces this over-coverage while lowering CRPS.

B. Role of Uncertainty Propagation and Feedback

The ablations indicate that propagating uncertainty into dispatch and closing the loop via feedback contribute more to objective reduction than rolling calibration alone. Intuitively, scenario MPC can hedge near constraints by choosing releases that are robust to inflow uncertainty, while adaptive inflation increases scenario spread when recent errors spike, providing a simple safeguard against temporary forecast degradation. Robustness experiments support this interpretation: under inflow scaling (Figure 8), feedback preserves violation rates while slightly reducing objective at higher α; under sparse/delayed observations (Figure 9), performance varies little from the nominal setting.

C. Operational Feasibility

The end-to-end runtime is compatible with daily decision-making. Figure 10 shows that most computation is spent in the dispatch optimizer; rolling calibration itself is negligible at the daily cadence. This suggests a practical scaling path: richer forecasters could be swapped into COMP-3, while maintaining the same closed-loop coupling and uncertainty-aware dispatch interface. Figure 11 summarizes the compute–value tradeoff: scenario-based pipelines incur higher runtime but remain in the sub-second-per-day regime, while delivering modest objective improvements.

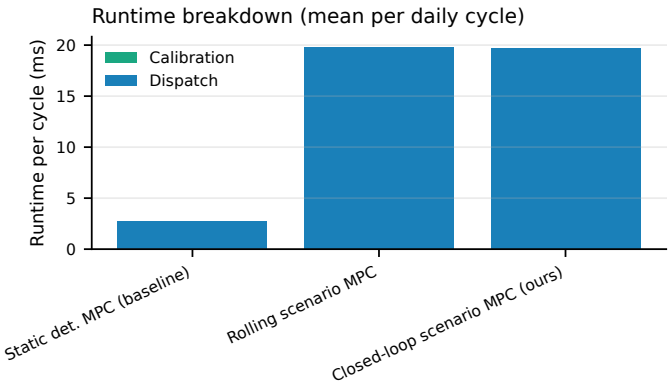

Fig. 10. Runtime breakdown per daily cycle into rolling-calibration and dispatch components (mean over reservoirs).

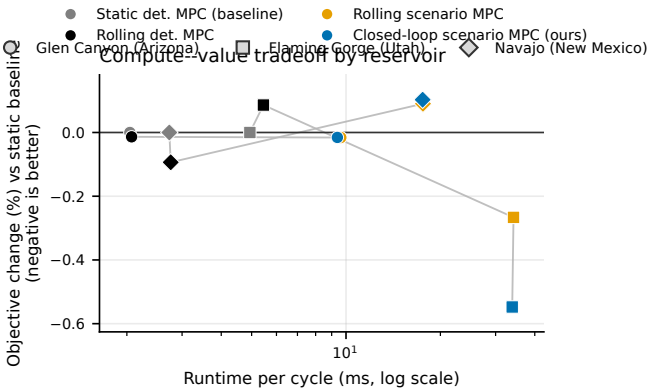

Fig. 11. Compute–value tradeoff across pipelines (test 2015–2020). Each point corresponds to one reservoir-method pair; color indicates method, marker shape indicates reservoir, and gray polylines connect the increasing-complexity pipeline within each reservoir. Runtime is shown on a log scale, and objective change is reported relative to the static deterministic baseline.

VII. LIMITATIONS

Several limitations bound the scope and generalizability of the present work.

Case-study dependence and data availability. The end-to-end value of the proposed framework is contingent on a reservoir system that provides sufficiently rich operational variables (storage trajectories, release schedules, and binding constraints) and objectives that admit reproducible specification [1], [2]. The mapping from forecast skill to operational value is inherently context- and regime-dependent [3], so the magnitude of improvements reported here should not be extrapolated to systems with substantially different hydrology or operating rules.

Limited statistical power and external validity. Evaluation is conducted on three reservoirs with complete daily records and a simplified objective-and-constraint proxy derived from historical operations. While this controlled setting supports systematic ablations and end-to-end traceability, it does not sustain strong claims of statistical significance.

Reported improvements in operational objectives are modest and reservoir-dependent. Establishing broader external validity across diverse reservoir types, objective formulations, and constraint specifications remains an open challenge [4].

Calibration stability under degraded observations. Rolling calibration and data-assimilation updates can become unstable when observations are sparse, biased, or temporally delayed. We explicitly probe this vulnerability through a targeted robustness experiment (EXP-ROB-OBS-SPARSE) and monitor stability diagnostics throughout all rollouts [5]–[7]; however, performance under more severely degraded observational regimes has not been characterized.

Uncertainty model approximation. The scenario generator adopts a Gaussian residual model with a $\sigma_t\sqrt{h}$ scaling heuristic to propagate uncertainty over an h-step horizon. This formulation neglects residual autocorrelation and model-structure mismatch, both of which can produce miscalibrated multi-step scenarios. More expressive uncertainty representations—such as non-Gaussian or temporally correlated error models—are not evaluated and may be necessary in systems with pronounced nonlinear dynamics.

VIII. Conclusion

We propose a closed-loop forecast–dispatch–implementation coupling framework with rolling calibration and uncertainty-aware dispatch. On three ResOpsUS reservoirs (test 2015–2020), rolling calibration improves probabilistic forecast accuracy (CRPS) relative to a static AR(p) baseline while keeping point-skill comparable. In end-to-end simulations, closed-loop scenario MPC with feedback yields the lowest average operational objective among the tested pipelines. Future work should evaluate richer objectives/constraints and additional reservoirs to characterize where closed-loop coupling provides the largest operational gains.

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